

# Network Traffic Analysis Using Wireshark

## Author

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## Tools Used

- Wireshark
- Windows 11
- Google Chrome
- Command Prompt

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## 1. Introduction

Network traffic analysis is a fundamental cybersecurity skill used to monitor, investigate, and troubleshoot communications between systems. By examining packet-level data, security professionals can identify normal network behavior, detect anomalies, and investigate potential security incidents.

This project demonstrates the use of Wireshark to capture and analyze network traffic generated during everyday internet activity. The analysis focuses on several core networking protocols, including DNS, TCP, HTTPS/TLS, and ICMP.

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## 2. Project Objectives

The primary objectives of this project were:

- Capture live network traffic using Wireshark.
- Analyze common networking protocols.
- Understand how devices communicate across a network.
- Identify security-relevant information contained within packet captures.
- Develop practical packet analysis skills used in cybersecurity operations.

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## 3. Lab Environment

### Hardware

- Windows 11 Workstation
- Wireless Network Connection

### Software

- Wireshark
- Google Chrome
- Windows Command Prompt

### Network Interface

- Active Wi-Fi Adapter

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## 4. Methodology

Network traffic was captured using Wireshark while performing common internet activities.

The following actions were conducted to generate traffic:

1. Opened multiple websites using Google Chrome.
2. Allowed DNS lookups to occur naturally during web browsing.

3. Established secure HTTPS connections to websites.
4. Generated ICMP traffic using the Windows ping command.
5. Captured packets and applied protocol-specific filters for analysis.

The following Wireshark display filters were used:

| Protocol  | Filter |
|-----------|--------|
| DNS       | dns    |
| TCP       | tcp    |
| TLS/HTTPS | tls    |
| ICMP      | icmp   |

## 5. DNS Analysis

### Overview

Domain Name System (DNS) traffic was observed whenever websites were accessed. DNS is responsible for translating human-readable domain names into IP addresses.

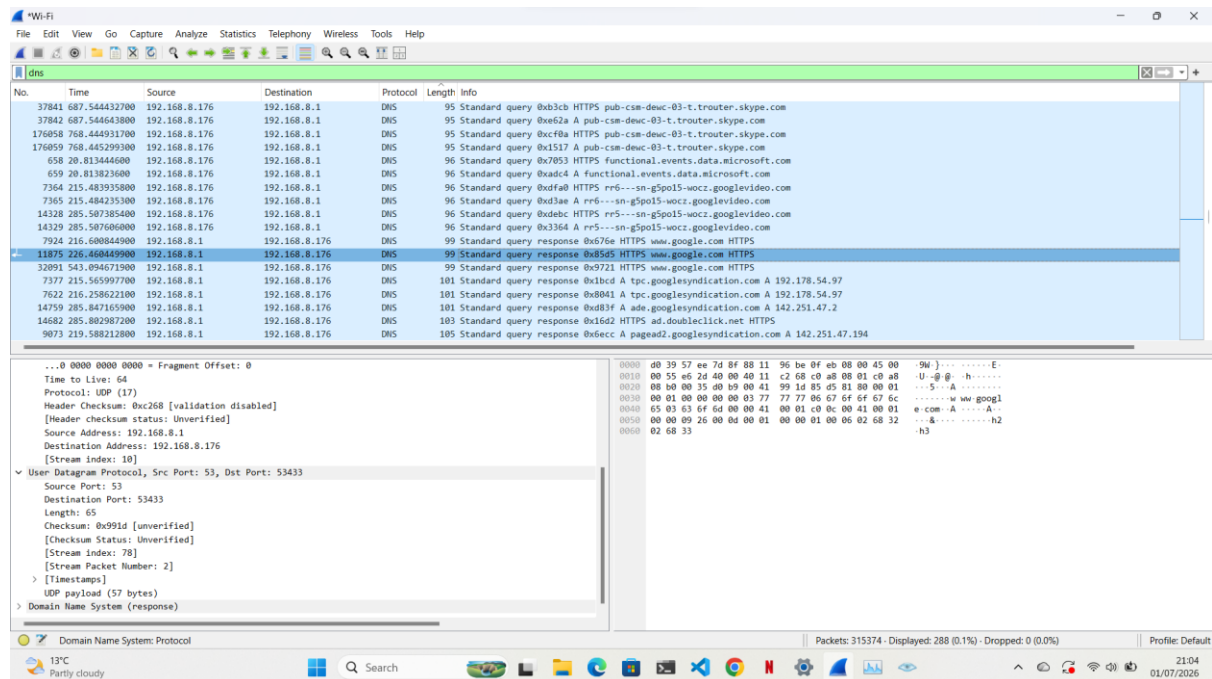
### Findings

- DNS queries were sent to the configured DNS server.
- Responses contained the IP addresses associated with requested domains.
- Multiple domain resolution requests were successfully completed.

### Security Relevance

DNS traffic can reveal websites visited by users and is often analyzed during security investigations and threat hunting activities.

### Screenshot



## 6. TCP Analysis

### Overview

Transmission Control Protocol (TCP) provides reliable communication between devices on a network.

### Findings

Multiple TCP three-way handshakes were identified.

The connection establishment process followed the standard sequence:

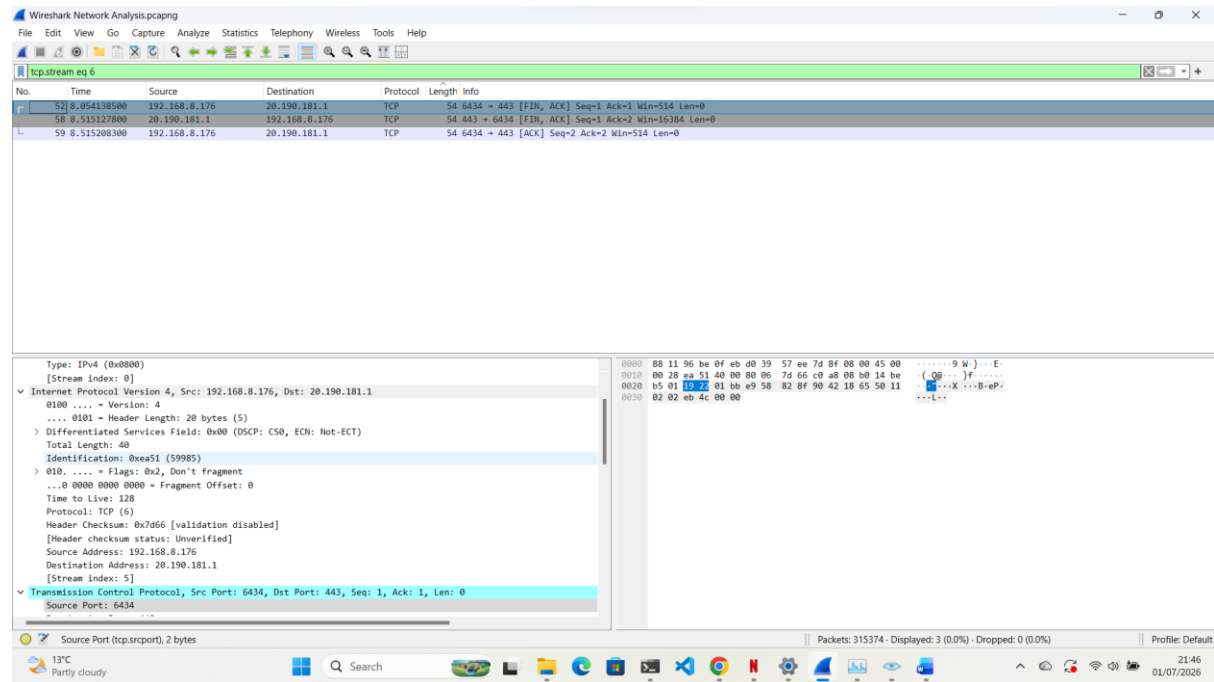
1. SYN
2. SYN-ACK
3. ACK

After successful connection establishment, data transfer occurred between the client and remote servers.

### Security Relevance

Monitoring TCP sessions helps analysts identify network communication patterns and troubleshoot connectivity issues.

### Screenshot



## 7. HTTPS/TLS Analysis

### Overview

Most modern web traffic is protected using HTTPS and Transport Layer Security (TLS).

### Findings

TLS handshakes were observed before encrypted communication sessions began.

The following activities were identified:

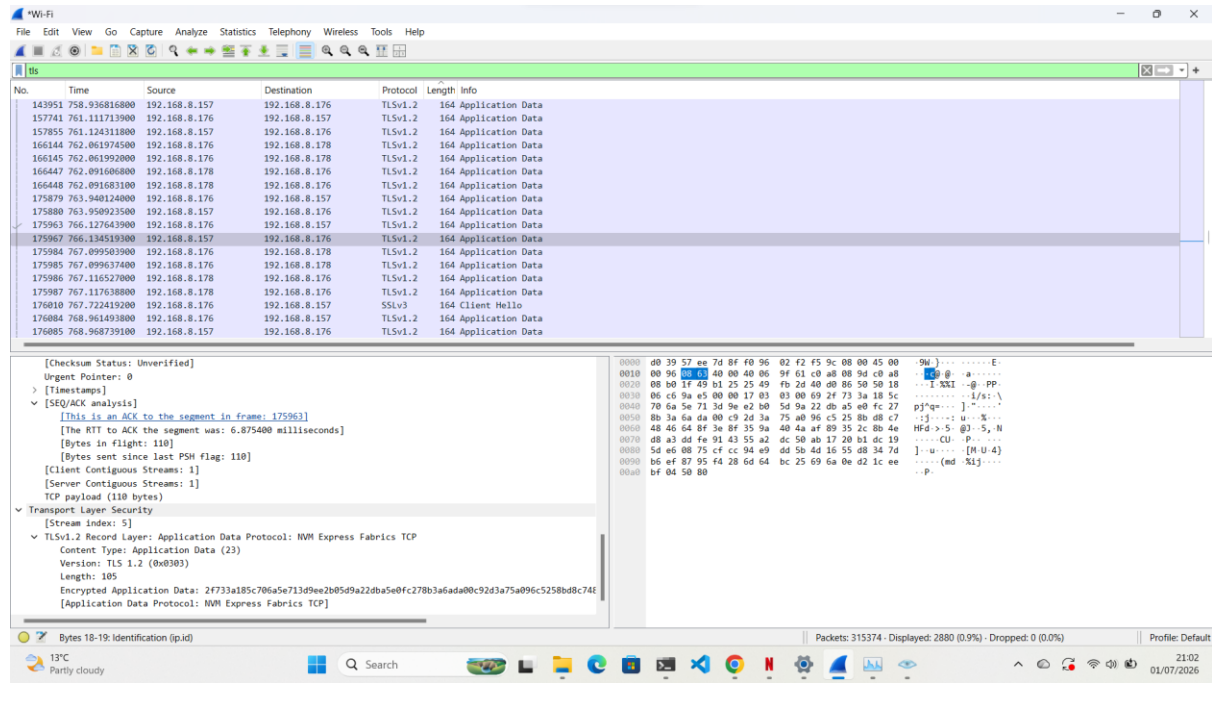
- Session initiation
- Certificate exchange
- Encryption negotiation
- Secure communication establishment

Application data remained encrypted after the TLS handshake.

### Security Relevance

TLS protects sensitive information from interception and is a critical component of modern cybersecurity.

## Screenshot



## 9. Security Observations

The following security observations were made during the packet analysis process:

- Most web traffic utilized TLS encryption.
- DNS traffic exposed visited domain names.
- TCP communications followed expected connection establishment procedures.
- ICMP responses confirmed network connectivity.
- No suspicious or malicious traffic was identified during the capture period.

These observations indicate normal network behavior throughout the capture session.

## 10. Skills Demonstrated

This project demonstrates practical experience in:

- Network Traffic Analysis
- Wireshark Packet Capture
- Protocol Analysis
- DNS Investigation
- TCP/IP Fundamentals
- HTTPS/TLS Inspection
- ICMP Troubleshooting
- Security Monitoring
- Cybersecurity Investigation Techniques

## 11. Lessons Learned

This project provided valuable hands-on experience with packet capture and protocol analysis.

Key lessons learned include:

- Understanding how DNS resolves domain names.
- Observing TCP connection establishment processes.

- Identifying secure communications through TLS.
- Using ICMP for network diagnostics.
- Applying Wireshark filters to efficiently analyze traffic.

These skills are directly applicable to Security Operations Center (SOC), network administration, and incident response roles.

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## **12. Conclusion**

Wireshark provides detailed visibility into network communications and remains one of the most important tools used by cybersecurity professionals.

Through the analysis of DNS, TCP, TLS, and ICMP traffic, this project demonstrated practical packet analysis techniques and reinforced foundational networking concepts. The knowledge gained through this exercise provides a strong basis for future work in cybersecurity monitoring, threat detection, incident response, and network troubleshooting.

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## **References**

- Wireshark Documentation
- RFC 791 – Internet Protocol (IP)
- RFC 793 – Transmission Control Protocol (TCP)
- RFC 1035 – Domain Name System (DNS)
- RFC 8446 – Transport Layer Security (TLS 1.3)